

Irish Math Olymipad 1989 — P9/10

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Let a be a positive real number, and let

$$b = \sqrt[3]{a + \sqrt{a^2 + 1}} + \sqrt[3]{a - \sqrt{a^2 + 1}}.$$

Prove that b is a positive integer, if and only if, a is a positive integer of the form $\frac{1}{2}n(n^2 + 3)$.

Solution

Let $x = \sqrt[3]{a + \sqrt{a^2 + 1}}$ and $y = \sqrt[3]{a - \sqrt{a^2 + 1}}$. Notice that $xy = \sqrt[3]{a^2 - (a^2 + 1)} = -1$.

Observe the following,

$$\begin{aligned} b^3 &= \left(a + \sqrt{a^2 + 1}\right) + 3xy(x + y) + \left(a - \sqrt{a^2 + 1}\right) \\ &= 2a + 3(-1)b \end{aligned}$$

and so $b^3 + 3b = 2a$ giving $a = \frac{1}{2}b(b^2 + 3)$ for some positive integer b .

Remark: This is a pretty dumb problem in that the most braindead approach (just cube both sides) actually works. It looks intimidating but is actually just really easy.